

Predicting hospital readmissions for diabetes patients

A Complete End to End Healthcare Analytics Project

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Learnings:

- Acquire and validate healthcare data
- Process messy real-world datasets
- Build predictive models
- Create professional visualizations
- Document everything on GitHub

Segment 1: Introduction s Data Acquisition

- Project overview and objectives
- Data source introduction (UCI)
- Environment setup (Python)
- Data download and initial loading

PROJECT OVERVIEW

Focus: End-to-end healthcare analytics project

Problem: Predicting 30-day hospital readmissions to reduce costs and improve patient outcomes

Tool: Python (exclusively)

Dataset: Diabetes 130-US Hospitals (1999-2008) - UCI Machine Learning Repository

Segment 2: Data Preprocessing s Cleaning

- Data quality assessment
- Missing value treatment
- Feature engineering
- Data transformation

Segment 3: Exploratory Data Analysis

- Univariate analysis
- Bivariate analysis
- Key insights identification
- Visualization creation

Segment 4: Exploratory Data Analysis

- Model building
- Model evaluation
- Feature Importance analysis

Segment 5: Reporting and Documentation

- Executive summary creation
- GitHub repository setup →
- Documentation best practices

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## 📁 Project Structure
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healthcare-readmission-prediction/
├── diabetic_data.csv           # Raw dataset (download separately)
├── diabetic_data_cleaned.csv  # Preprocessed dataset
├── healthcare_analytics.py     # Main analysis script
├── EXECUTIVE_SUMMARY.txt      # Executive summary report
├── README.md                  # This file
├── visualizations/           # All generated visualizations
│   ├── 01_target_distribution.png
│   ├── 02_numeric_distributions.png
│   ├── 03_categorical_distributions.png
│   ├── 04_numeric_vs_readmission.png
│   ├── 05_categorical_vs_readmission.png
│   ├── 06_correlation_matrix.png
│   ├── 07_model_comparison.png
│   ├── 08_confusion_matrices.png
│   ├── 09_feature_importance.png
│   ├── 10_probability_distribution.png
│   ├── 11_risk_stratification.png
│   └── 12_FINAL_DASHBOARD.png
└── requirements.txt           # Python dependencies
...
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